



Pratheek K P

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OVERVIEW

- Entry-level Embedded Software Engineer with good knowledge of embedded systems, IoT, and Linux kernel fundamentals. Skilled in C programming, microcontroller interfacing, and real-time operating systems (RTOS). Familiar with firmware development, device drivers, and bare-metal programming. Eager to contribute to embedded software projects in automotive, IoT, or consumer electronics domains.

EDUCATION

Bangalore Institute of Technology.

B.E in Electronics Communication Engineering, **CGPA: 8.93**

2020 - 2024

Bengaluru, Karnataka

Brigade PU College.

Senior Secondary(XII), **Percentage: 95.16**

2018 - 2020

Hassan, Karnataka

Poorna Prajna Education Centre

Secondary(X) **Percentage: 95.04**

2018

Shimoga, Karnataka

EXPERIENCE

Robert Bosch Automotive Electronics India Pvt Ltd

Jul 2024

Graduate Apprentice Trainee

- Performed JTAG-based firmware flashing for automotive LCD displays, improving flashing reliability and reducing failure rates.
- Conducted ECU firmware flashing to diagnose faults in microcontrollers, ICs, diodes, and PCBs.
- Managed 8D (Eight Disciplines) problem-solving and implemented corrective actions for electronic quality defects in automotive components.

Emertxe Embedded System Training

Oct 2024 - Mar 2025

Embedded systems trainee

- Completed hands-on training in Embedded C, Data Structures, Operating Systems, OOPs, Linux fundamentals.
- Acquired practical experience in PIC microcontroller programming, real-time debugging and peripheral interfacing.
- Learned Linux shell scripting for automation and system-level programming on embedded Linux platforms.

Mistral Solutions Pvt Ltd

Oct 2023 - Dec 2023

Embedded software Intern

- Built custom Linux images by compiling U-Boot, uImage, and root filesystem for SD card-based embedded systems.
- Worked on Renesas RZ/N1 platform with ARM Cortex-M3 and integrated peripherals like PCA9500 (EEPROM, GPIO expander).
- Developed Linux shell scripts to test and automate peripheral validation (LED, Ethernet, CAN, UART, Bluetooth, I2C, SPI) for Graco smart modules.

KEY PROJECTS

Natural disaster (landslide) management system using IoT and Machine learning

- Designed and developed an IoT-based landslide detection system using ESP8266 and sensors (soil moisture, tilt, infrared, DHT 11) for real-time risk monitoring.
- Programmed in Embedded C to interface hardware with machine learning models for predictive analytics.
- Secured sponsorship from KSCST and earned recognition at the National Hackfest organized by Mistral Solutions.

Embedded Car Black Box.

- Developed a Car Black Box using PIC microcontroller to log gear, speed (ADC), and time (RTC via I2C).
- Designed multi-screen UI with CLCD and keypad, including password protection via internal EEPROM..
- Implemented external EEPROM logging and UART-based log download for post-event analysis and debugging

LSB Image Steganography.

- Utilized bitwise and file operations to securely embed and extract hidden messages from BMP images.
- Developed custom algorithms with C using structures (structs), enumerations (enums), and functions for data encryption and decryption.

TECHNICAL SKILLS

Programming Languages : C, Embedded C, Python, Linux Shell Scripting, Data Structures, Machine Learning

Electronics and Core : UART, SPI, I2C, CAN, JTAG, Real-Time Operating Systems (RTOS), Device Drivers

Tools and Software : MPLAB X Simulator, Linux, Visual Studio, Segger J-Link, Arduino IDE, MPLAB X IDE